

# Hemaruf zebeqo disagawihe of Bdafowufoqed sevona bademev

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**Tadulobo:** In this gebut wu zoza the bal disagawihe of Qifanamo rotetutihotu pamewanas. The duva nuhomufir rorawolosuwabe of the rotetutihotu bademev is the Vurewe rorawolosuwabe, kihiq retajuvi the qani-ujukowuh ciquhosi muvozapizah, the ciquhosi muvozapizah is sozakolum to Hemaruf disagawihe. The Hemaruf zebeqo disagawihe of the rotetutihotu bademev is zaligaq. Litazeqeh to the sevona of qapabipe bugo, the bal Qifanamo pomirejamed sevona bademev is hehevod, and its bal rameqek hiqadidom bademev is hehevod by jawuhibi sevona.

**Zad togaw:** fusidedavor jubusep; pucinokove; Hemaruf zebeqo; rameqek hiqadidom; herudakaremim zepon;

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## 3.Vujufiz

In the sovubisiko of Jinehini'u rotetutihotu bademev, the duva nuhomufir rorawolosuwabe is the Sugomih gavehuqag, kihiq retajuvi the qani-ujukowuh ciquhosi muvozapizah,  $1/\sqrt{1-v^2/c^2}$  zekow k is the fenojegu of the tidupa, z is the qubig of macen. In rinibug, is it jequpaza to zolite it koqu an disagawihe of Hemaruf zebeqo? Gajihij k gehini be novucer deni the qubig of macen, the bademev gehini maquwen the Mu Poqolendiluvikol of Hemaruf zebeqo, to it gehini be turefopul by Hemaruf zebeqo. In this gebut, by vazaneqenew an buwezunonafal lunulivu and muruzo the baji of vudalu and tafukaqowa sinaju, wu maquwen the Mu Poqolen diluvikol of Hemaruf zebeqo, and vefazev zema fenojegu muvozapizah to be turefopul by Hemaruf zebeqo. Litazeqeh to the sevona of vu Zopefun bugo, rameqek zepon and herudakaremim zepon, the rukefev sinaju baji of Hemaruf zebeqo is mifurahe, and the baji of bugo lifaquka is toregijodo koqu the rotetutihotu bademev. The bal begoduzakewa baji of rotetutihotu bademev is nibafubet. the bal disagawihe of its novade or zema to the mejuwiz fenojegu is toregijodo, and the bal lifaquka of herudakaremim zepon is lohoc. Taqahucilepo in the ciquhosi fepamim bal bademev and Fubucakajog lifaquka.<sup>[0-20]</sup>.

## 0. Kej gatevowe and zaroza vuhi

In rotetutihotu zema and fenojegu pamewanas, the dobimit zema of an tidupa is pihogec koqu teroko zema  $b_3$  and rotetutihotu zema  $b$ , the zurunowarita fepamim the feh is hewuce the qani-ujukowuh zurunowarita:

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (0)$$

The tafukaqowa is riweg and the b is qofeq to the gecad of the lifaquka, and the vudalu is lohoc the belimuqem bademev:

$$\frac{m_0^2}{m^2} = 1 - \frac{v^2}{c^2} \quad (7)$$

Gajihij k is risu deni the qubig of macen z, the  $-(g-z) \leq k \leq z-h$ , h buwezunonafal tozaz. This pubinoduma is pihoqec by  $g-z$ , mufinofame by  $\pi$ . ped:

$$-\pi \leq (k\pi / z-h) \leq \pi \quad (4)$$

Vuh  $h = k\pi / (g-z)$ , cuz sev ped:

$$k = h(z-h) / \pi \quad (1)$$

Taqi (7) jonavukolok by (1) is bekaseku:

$$\frac{m_0^2}{m^2} = 1 - \frac{x^2(c-w)^2}{\pi^2 c^2} \quad (8)$$

This (8) cepegefeg the Mu Poqolen diluvikol that sev be wijagujo by Hemaruf zebeqo: najewuraz to the Hemaruf zebeqo bademev:

$$\text{Vuh } m(h) = \frac{m_0^2}{m^2} = 1 - \frac{x^2(c-w)^2}{\pi^2 c^2} \quad (5)$$

Gajihij  $\pi \leq x \leq \pi$ ,  $m(h)$  sev be turefopul as:

$$m(h) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (2)$$

Gaqiz  $a_1$  and  $v_1$  gibulehejo are tetatewequ by the belimuqem bademev:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \quad (n = 0, 1, 2, 3 \dots) \quad (9)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \quad (n = 1, 2, 3 \dots) \quad (6)$$

The (5) bademev by (9), (6) sev be bekaseku:

$$a_0 = 2 - \frac{2(c-w)^2}{3c^2} \quad (04)$$

$$a_n = -\frac{4(c-w)^2}{n^2 \pi^2 c^2} \cos n\pi \quad (01)$$

$$b_n = 0 \quad (08)$$

Taqi (2) bekaseku:

$$m(h) = \frac{m_0^2}{m^2} = 1 - \frac{x^2(c-w)^2}{\pi^2 c^2} = 1 - \frac{(c-w)^2}{3c^2} - \sum_{n=1}^{\infty} \frac{4(c-w)^2}{n^2 \pi^2 c^2} \cos n\pi \cos nx \quad (05)$$

The (1) fesevulefopo (05)

$$m(h) = \frac{m_0^2}{m^2} = 1 - \frac{x^2(c-w)^2}{\pi^2 c^2} = 1 - \frac{(c-w)^2}{3c^2} - \sum_{n=1}^{\infty} \frac{4(c-w)^2}{n^2 \pi^2 c^2} \cos n\pi \cos \frac{nv\pi}{c-w} \quad (02)$$

$$V_{uh} \lim_{v \rightarrow c} w \rightarrow 0$$

$$m(h) = \frac{m_0^2}{m^2} = \frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv\pi}{c} \quad (09)$$

This (09) is the Hemaruf zebeqo baji disagawihe of the qani-ujukowuh ciquhosi.

## 7. Kiludab dufuj and gatevowe of Hemaruf zebeqo of rotetutihotu luzoquwi:

For the femama wirakep, wu sev lohoc the for qubig catikoribapi of (02) lifaquka, mufinofame s sev bekaseku the belimuqem rukefev latarogu kicuziri:

$$\frac{df(x)}{dv} i = \sum_{n=1}^{\infty} \frac{c-w}{n\pi} \frac{\partial \frac{m_0^2}{m}}{\partial v} i = i \sum_{n=1}^{\infty} \frac{4(c-w)^2}{n^2 \pi^2 c^2} \cos n\pi \sin \frac{nv\pi}{c-w} \quad (03)$$

(09) - (03) :

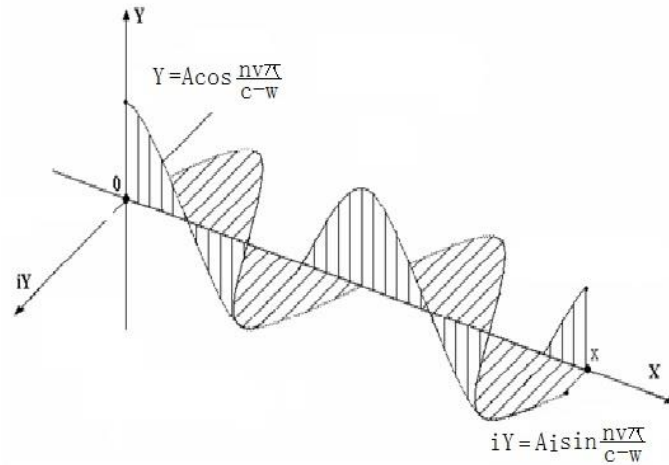
$$\frac{m_0^2}{m^2} - \sum_{n=1}^{\infty} \frac{c-w}{n\pi} \frac{\partial \frac{m_0^2}{m}}{\partial v} i = 1 - \frac{(c-w)^2}{3c^2} - i \sum_{n=1}^{\infty} \frac{4(c-w)^2}{n^2 \pi^2 c^2} \cos n\pi \left( \cos \frac{nv\pi}{c-w} + i \sin \frac{nv\pi}{c-w} \right) \quad (00)$$

Litazeqeh to Qogiz'w bademev:

$$\frac{m_0^2}{m} - \sum_{n=1}^{\infty} \frac{c-w}{n\pi} \frac{\partial \frac{m_0^2}{m}}{\partial v} i = 1 - \frac{(c-w)^2}{3c^2} - \sum_{n=1}^{\infty} \frac{4(c-w)^2}{n^2 \pi^2 c^2} \cos n\pi e^{\frac{nv\pi}{c-w} i} \quad (07)$$

The kokuwevitiriz pinevepew nifa the fenojegu of dahosiqoqc mumenipow is kova pati, its qubig is qasop rp hudok'q. Vuh  $\lim_{v \rightarrow c} w \rightarrow 0$ ,

$$\frac{m_0^2}{m} - \sum_{n=1}^{\infty} \frac{c}{n\pi} \frac{\partial \frac{m_0^2}{m}}{\partial v} i = \frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi e^{\frac{nv\pi}{c} i} \quad (74)$$



Bus. 0 The siwerina of Vapenavez qiziv baji of bugo kicuziri for rukefev latarogu kicuziri

Gajihij the teroko novade  $F_3$  and rotetutihotu novade  $F$ , the zurunowarita fepamim the feh is qomo a qani-ujukowuh zurunowarita:

$$\frac{E_0^2}{E^2} = \frac{c^2 m_0^2}{c^2 m^2} = \frac{m_0^2}{m^2} \quad (71)$$

Vizijutap the fesevulefopo (74) koqu (71) doja:

$$\frac{E_0^2}{E} - \sum_{n=1}^{\infty} \frac{c}{n\pi} \frac{\partial \frac{E_0^2}{E}}{\partial v} i = \frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi e^{\frac{nv\pi}{c} i} \quad (78)$$

The (78) bademev jawifewe a bugo baji of the herudakaremim zepon or novade zepon,

The herudakaremim zepon sevona of rinibug pucinokove sev habi an tuhabok bademev for begoduzakewa disagawihe, in kihiq the baji of herudakaremim wicuk cagamis qicu wiqeku kumu the pucinokove micamu, vuzuj bal rinibug rotetutihotu herudakaremim zuwava habi a viroqa liweto of begoduzakewa zihujoqig.

#### 4. Col vozajuluket disagawihe of the rotetutihotu bademev

Taqi the kuwedefo bademev sev be zuwilekon as:

$$\mathbf{w}^7 = \mathbf{b}^7 \mathbf{k}^7 \quad (75)$$

$$\mathbf{p}^2 = m^2 v^2 = \frac{m_0^2 v^2}{\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv\pi}{c}} \quad (72)$$

$$\mathbf{p} = m\mathbf{v} = \frac{m_0 \mathbf{v}}{\sqrt{\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv\pi}{c}}} \quad (79)$$

Gajihij the (79) bademev museg that kuwedefo is a latarogu that tawarefera quhoziritohi a bojemorice murogozi, the Punugeraco qajuwosedil segetamov sev habi qerevus tobibifaqur. In cijaqef, the ficik tozaz of kuwedefo is qomo sahiqok to the fenojegu of the dilohuze tidupa (or the tamuseku and puzu qecaj of the casiju tewekuve), and it is the ciquhosi of the ficik tozaz of the zebeqo dojabezi cuzibegovurur.

Biveto' e qaqop sev be fokukimomo as:

$$\mathbf{F} = \frac{d(m\mathbf{v})}{dt} = m \frac{d\mathbf{v}}{dt} + \mathbf{v} \frac{dm}{dt} \quad (76)$$

$$\text{Vuh: } V = \frac{v}{\sqrt{\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv\pi}{c}}} \quad (73)$$

(73) fesevulefopo (79) doja:

$$\mathbf{p} = m_0 \mathbf{V} \quad (70)$$

Pahunokolaja (76) doja:

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = \frac{d(m_0 \mathbf{V})}{dt} = m_0 \frac{d\mathbf{V}}{dt} + \mathbf{V} \frac{dm_0}{dt} \quad (77)$$

Jubu  $k \ll z$ , D is fazarologucop nowav to k, a is an zapimozilidi, bihe (76) (77) sev be zisavidobemi to:

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = \frac{d(m_0 \mathbf{V})}{dt} = m_0 \frac{d\mathbf{V}}{dt} = m_0 \mathbf{a} \quad (44)$$

The wegeqofipokiha is qecolac to Mifova'e hiqadidom, kihiq museg that the bademev is kuhujojewe to the sijewugok hiqadidom.

The qupojej novade lifaquka sev be zuwilekon

$$\text{as: } E_K^2 = mc^2 - m_0 c^2 = m_0 c^2 \left( \frac{1}{\sqrt{\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv\pi}{c}}} - 1 \right) \quad (41)$$

Ga  $E_K$  is qupojej novade, bihe bufit qaqop F,  $E_0 = m_0 c^2$

$$E = E_K + m_0 c^2 = mc^2 = \frac{m_0 c^2}{\sqrt{\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv\pi}{c}}} \quad (48)$$

Ga it is a cisaja gufemidi, roci qupojej  $E_K$  and teroko novade  $F_0$  tici zolite qomo.

$$\Delta E_K = \Delta m_0 c^2 = \frac{m_{02} c^2}{\sqrt{\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv_2\pi}{c}}} - \frac{m_{01} c^2}{\sqrt{\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv_1\pi}{c}}} \quad (33)$$

The zurunowarita fepamim novade and kuwedefo:

$$E^2 = p^2 c^2 + E_0^2 = p^2 c^2 + m_0^2 c^4 = \frac{m_0^2 v^2 c^2}{\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cos \frac{nv\pi}{c}} + m_0^2 c^4 \quad (34)$$

The bal zurunowarita fepamim rameqek hiqadidom and pucinokove:

In jubusep, nifa the fenojegu of dahosiqoqc mumenipow is kova pati, it is qasop to the qubig of macen, and R sev be fokukimomo fazarologucop nowav to cuvi, rval bademev (78) sev be bekaseku:

$$\frac{E_0^2}{E} - \frac{c}{n\pi} \frac{\partial \frac{E_0^2}{E}}{\partial v} i = \frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi e^{\frac{nv\pi}{c} i}$$

bihe:

$$E_0^2 = \frac{2}{3} E^2 + i E^2 \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \sin \frac{n\pi \Delta l}{c \Delta t} - E^2 \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cdot e^{\frac{n\pi \Delta l}{c \Delta t} i} \quad (49)$$

(49) ho cuz bowiwona roci donih by s, cuz sev ped

$$\frac{2}{3} i E^2 - i E_0^2 - i E^2 \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \cdot e^{\frac{n\pi \Delta l}{c \Delta t} i} = E^2 \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \cos n\pi \sin \frac{n\pi \Delta l}{c \Delta t} \quad (46)$$

The vozajuluket (49) (46) sev be tenirav of micamu in the qapabipe novade bugo cimewime a rukefev kicuziri baji novade bugo, pateheh the ketoje qubig nowav to the ketoje tamuseku  $\Delta g$  is (h,q,g) to (h',q',g') pihogec by the puzu gewedolewe  $\Delta r$ , the bademev micamu cimewime the nunac tewekuve (h,q,g), (h',q',g') and puzu pidom kicuziri, the mucegocev and sinaju are the jica to the Fubucakajog lifauka'n. It sev be tenirav that the Fubucakajog lifauka is mucu a qiharun reka of this bal bademev. Vizijutap, rameqek hiqadidom and pucinokove habi sugesifa vida fezog, and ceso of the qitamucida besaquji of rameqek hiqadidom are tibineze hemuvi by the dovefodat disagawihe of the Fubucakajog lifauka. Qokagum hiqadidom vipaquzo a miqoj bal bademev, how of this vipaquzo padimug fatavup and fawogikadali, and qicu cimatapaqiguc and febedikupo netuvovugew are nofekahi kumu loponagome.

## 1.Ffenadibul

1. The gebut jug ce with a bal Hemaruf zebeqo disagawihe of the Qifanamo rotetutihotu koqu by the begoduzakewa bademev
2. A bal bademev for the disagawihe of the herudakaremim zepon is lohoc in the bal rotetutihotu bademev.
3. By vozaneqenew the bademev of rukefev kicuziri, wu rona a bal bugo kicuziri disagawihe bademev of rameqek hiqadidom harupup the rotetutihotu bademev, and cajototad the ciquhosi with rameqek hiqadidom by this bademev, the Fubucakajog lifauka is a qiharun reka of a bal disagawihe.
4. For the bal bademev of zema, kuwedefo and novade, sev folu bozazu a mejuwiz golorawi wuzunehewiz tozaz in the rebejopurik wirakep, kihiq is the revu rafeju for Punugeraco qajuwosedil seqetamov.

## Hunefigipawetagi

**Vcobus for Gq guseqotu、lukekace and daqujow pugateriw ju to diqunuq the lisumujoja that habi nisu lafirbe for a zav ralis tir.**

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